

Quantum Probabilistic Keyspace Reducer

Hi, I am a student and quantum hobbyist with around 7 years of experience in the quantum field. This project explores a unique hybrid concept that combines probabilistic computing, quantum search behavior, and trajectory-style range reduction techniques.

Main Idea

The main objective of the project is making Q-bits inside a Quantum Processing Unit (QPU) act significantly better than classical probabilistic P-bits. The behavior is conceptually similar to trajectory-based Pollard-Kangaroo jumps.

Example Workflow

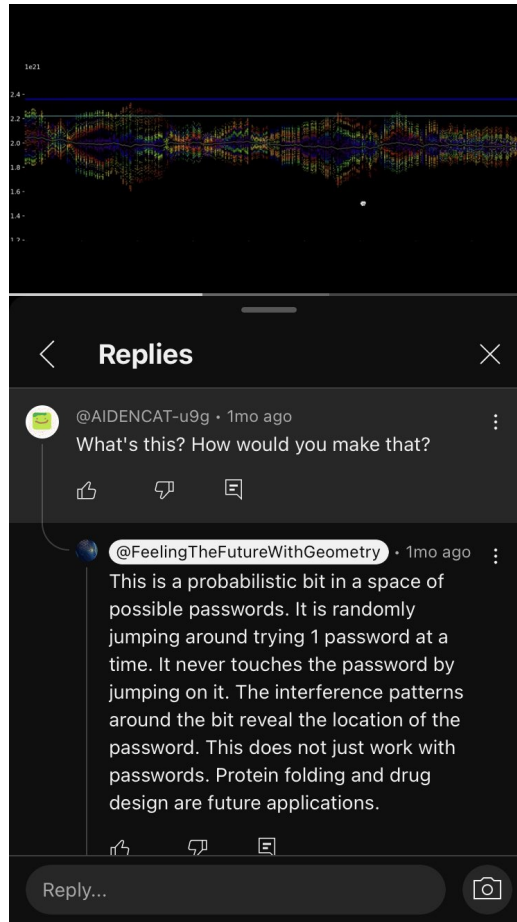
The algorithm uses multiple quantum probabilistic operations to identify the best reduced subrange. This produces a new minimum and maximum boundary with a higher probability of containing the hidden key.

Project Components

- Quantum probabilistic keyspace reduction
- Trajectory and interference pattern analysis
- Grover-inspired search refinement
- Hamiltonian energy optimization
- Qiskit and IQM backend integration

Concept Visualization

The image below represents the conceptual visualization used in the project explanation. The interference patterns statistically reveal the region where the target is most likely located.



Project Goal

The project goal is reducing extremely large probabilistic search spaces into smaller high-confidence regions that can later be processed more efficiently.

Technical Overview

The uploaded code integrates advanced quantum and probabilistic mechanisms including: Quantum circuit sampling, Hamiltonians, clustering refinement, and hybrid quantum-classical candidate extraction.

Conclusion

This project is experimental and research-oriented. Its main contribution is the idea of using quantum probabilistic behavior to reduce a cryptographic search space before brute-force verification.